COMSATS University Islamabad, Lahore Campus

Department of Electrical and Computer Engineering

Mid-Term—Semester Spring 2024

Course Title:	Game Development				Course Code:	CSC495	Credit	3(2,1)&
Course Instructor/s:	Saira Aslam				Program Name:	BCE/BCS/BSE		
Semester:	8th and 6th	Batch:	FA20/FA21	Section:	Elective	Date:	25th April 2024	
Time	**1 hour 30 minutes**				**Maximum Marks:**		**50**	
Student's	[redacted]				Reg. No.	[redacted]		

Important Instructions / Guidelines:

- Total number of pages are 4
- Make sure to write your Name and Registration number at the top of Question paper and Answer sheet
- Write Answer of Question 1 and 2 on Answer sheet
- Solve Q3 and Q4 on Question paper
- Do not exceed the space limit

Question No 1. **Marks: 5+5 = 10**

CLO: <2>; Bloom Taxonomy Level: <Understand>

a- How will you instantiate 10 cars in a line (in z-axis) in Unity Game scene when game starts?
Hint: Write code which has **Vector3 pos = new Vector3(___??_____)** inside a loop and **Instantiate** built-in method in a script which is attached to empty game object

b- Explain how bounciness can be added to a sphere in Unity editor?

Question No 2. **Marks: 4+4+4 = 12**

CLO: <2>; Bloom Taxonomy Level: <Understand>

Create an Animation Controller showing:

1- all **states**
2- all **parameters**,

that will make a character walk in start and die when hit by bullet keeping in view that you already have both animation clips, i.e., "walk" and "die,

3- Also write a method which makes the character die when hit by a bullet.

Question No 3.

15

CLO: <4>; Bloom Taxonomy Level: <Analyze> **Marks: 2x10 = 20**

Analyze the given C# Scripts (for Unity3D/2D game) below, and answer the following questions in the space provided below in the question paper:

a-	**public class Move : MonoBehaviour{** **void Start()** **{** **Transform.Translate(0, 0, 1.0f);** **}** **}** if this script is attached to a 3D bus, for how long bus will move?	Answers: Bus will move only when it becomes active and run 1.0f toward z-axis only once (one time).

b-	**private void OnCollisionEnter(Collision col)**
	{
	if(col.gameObject.name.StartsWith("enemy"))
	{
	Destroy(col.gameObject); //→ 1st IF STATEMENT
	}
	Destroy(transform.gameObject); //→ 2nd IF STATEMENT
	}
	
	If the above script is added to a bullet in a 3D game, which objects will be destroyed in both IF statements?
		First IF: GameObject collided with bullet will be destroyed having tag "enemy".
		Second IF: GameObject to which script is attached will be destroyed. It means bullet will be destroyed.

c-	**public class coinController : MonoBehaviour**
	{
	public GameObject coins;
	void Start() {
	for(int i = 0; i < 5; i=i+2){
	
	Vector3 CoinPos = new Vector3(Random.Range(1, 70), Random.Range(-4, 4), Random.Range(0, 50);
	
	Instantiate(coins, CoinPos, transform.rotation);
	}//end for
	} // end start()
	}//end class
	
	If the above script is added to an empty game object, how many coins are instantiated in the game through it?
		No. of coins: 3

d-	In the same above script (part c) what could	

	be value of **CoinPos.z** on every iteration?	Value of CoinPos.z = Random.Range (0,50); It will be random on every iteration b/w 0 and 50.
e-	**if (Input.GetKey(KeyCode.RightArrow)){** **Transform.Translate(0, 0, 1.0f);** **}** **if (Input.GetKeyDown(KeyCode.LeftArrow)){** **Transform.Rotate(0 ,0, -1.0f);** **}** If the script is added to a 3D character, in what direction player will move, and what difference can be seen in the game for two types of given inputs?	If player pressed ~~left~~ right arrow key, 3D character to which script is attached will move in forward direction. If "left arrow" key is pressed, game object will move in backward direction due to -ve z-axis value.
f	**public static bool muteMyGame = false;** **public void muteGame()** **{** **gameAudio.mute = true;** **muteMyGame = true;** **MuteButton.gameObject.SetActive(false);** **UnMuteButton.gameObject.SetActive(true);** **}** If a game is set to mute by calling above method on a MuteButton in one scene of a game, which variable will tend to keep it mute in the other scene of the same game? And how will it work?	AudioListener.volume = 0;

g	**public void shoot(){**
	RaycastHit hit;
	if (Physics.Raycast(fpsCam.transform.position, fpsCam.transform.forward, out hit, range))
	{
	??????????
	}
	}
	What will you add inside above if statement to destroy game object named as enemy?
		if(hit.transform.tag == "enemy")
		{
		Enemy enemy = hit.transform.GetComponent<Enemy>();
		if (enemy != null){
		enemy.die();
		}}

h	**if (Input.GetKeyDown(KeyCode.Space))//instantiate bullet from fighter jet** **{** **Vector3 jet_position = transform.position;** **??**	jet_position.y -= 1f;

	Instantiate(bullet, jet_position, transform.rotation); } What changes are required in jet_position to instantiate the bullet one point below the fighter jet?	

i	if (Input.GetKey(KeyCode.RightArrow))
	{
	transform.Translate(~~0.5f,0~~ 0.5f,0,0);
	}
	Robot
	if (Input.GetKey(KeyCode.LeftArrow))
	{
	transform.Translate(-0.5f,0,0);
	}
	If the above script is attached with the car, what method should be called in blank space to turn the car left/right respectively.
		0

j	public class AnyScript : MonoBehaviour
	{
	public static float scoreValue=0;
	public Text scoreText;
	
	void Start()
	{
	scoreText.text = "Score: " + scoreValue.ToString();
	}
	}
	
	What will be the value of **Score:** when play mode starts?
	What is the role of ToString() method in above code?
		1) Value of score will be '0' (zero).
		2) To convert float or integer to string so that two strings are concatinated because 'scorevalue' is not string but float. So, we need typecasting. Although it does not generate error in current version of unity.

8 **Marks: 4+4**

Question No 4.

CLO: <4>; Bloom Taxonomy Level: <Analyze>

While making any 3D game, how following problems can be resolved in Unity:

1- If a 3D human character is sinking in Mountains of a terrain while walking

We need to attach Rigidbody and Collider (static) to the mountain to simulate physics and turn gravity "off". 4

2- If a 3D human character is rotating unnaturally when hit by a bullet

We need to ~~control~~ freeze its x, y or z axis according to the role of character in game by choosing check boxes in "Constraints" of rigid body. 4